Exam 2 — Implementation and failure diagnosis

Questions only

Name: _____ **Date:** _____

Conditions

- 50 questions: 30 single-answer multiple choice and 20 numerical. Each is worth 1 raw practice point.
- Suggested training limit: **150 minutes**. This is not a claim about the actual exam duration or question count.
- Use up to three double-sided A4 sheets, prepared by you, and a local Jupyter notebook. No internet, AI assistance or answer key during the attempt.
- Use Python/NumPy for computation; SciPy is also allowed. You do not need executable OpenCV or PyTorch. Specialized routines are provided when needed.
- For multiple choice, select exactly one of A–D. Attempt every question.
- Unless an exact integer is requested, round the final numerical answer to **two decimal places**. Keep full precision in intermediate steps. Use fractions rather than percentages unless specified.
- Numerical practice grading uses the syllabus rule: within ± 0.01 of the unrounded reference, inclusive. Counts require the exact integer; no partial credit.
- Conventions stated in a question override defaults. Arrays use [row,column] = [y,x]; geometric vectors are columns; image-plane projection is virtual/front-facing; foreground is 1.
- This is original, challenging training material based on the supplied course files, not a lecturer-issued mock or a prediction of the exam.

Solutions are in a separate PDF. Do not open them until you finish.

Part A — Single-answer multiple choice

Q01 — Multiple choice

R_1 and R_2 are proper 3D rotation matrices. Which statement is correct?

A. The inverse is always $R_2^\top R_1^\top$.

B. A rotation matrix has 9 independent degrees of freedom.

C. The inverse of $R_2 R_1$ is $R_1^\top R_2^\top$; rotation order generally matters.

D. $R_1 R_2$ always equals $R_2 R_1$.

Q02 — Multiple choice

Crop an image by removing 40 pixels from the left and 20 from the top, with no resize. In the continuous-coordinate model, how does K change?

A. Subtract 40 from f_x and 20 from f_y.

B. Keep K unchanged and add $\begin{bmatrix}40 & 20 & 0\end{bmatrix}$ to camera translation.

C. Subtract 40 from c_x and 20 from c_y; keep focal lengths and skew unchanged.

D. Multiply both focal lengths by the crop's remaining image area.

Q03 — Multiple choice

For floating-point RGB image I of shape (H,W,3), what does this code compute?

```
out = (I.max(axis=2) + I.min(axis=2)) / 2
```

A. One scalar containing the midpoint of the global intensity range.

B. A lightness-based grayscale image of shape (H,W).

C. An RGB image with corrected white balance.

D. A hue map obtained by averaging the red and blue channels.

Q04 — Multiple choice

Which expression computes per-channel image means that broadcast correctly onto an array of shape (H,W,3)?

A. `I.mean(axis=0, keepdims=True), which always has shape (1,1,3)`

B. `I.mean(axis=(0,1), keepdims=True)`

C. `I.mean(axis=(0,1,2), keepdims=True)`

D. `I.mean(axis=2, keepdims=True)`

Q05 — Multiple choice

Suppose the true lighting matrix is $L_{\text{true}} = 2L_{\text{used}}$, with no shadows or noise. You solve $i = L_{\text{used}}\hat{g}$. How do the estimated normal and albedo relate to the true ones?

A. The normal doubles in length after normalization; albedo is unchanged.

B. Both normal direction and albedo are unchanged.

C. The normal reverses and albedo is halved.

D. The normal is unchanged; the albedo estimate is doubled.

Q06 — Multiple choice

For the intrinsic-image model $I = A \odot S$, where multiplication is elementwise, which recolouring operation preserves the estimated shading?

A. Divide the target albedo by S.

B. Change the object's albedo A and multiply the changed A by the original S.

C. Change S to the target RGB colour and discard A.

D. Replace every RGB pixel in I by the same target colour, including shadows.

Q07 — Multiple choice

A uniformly painted matte object has spatially constant albedo. Why can its image still contain a brightness gradient?

A. Lambertian reflection requires all pixels to have the same intensity.

B. Constant albedo forces constant depth.

C. Its surface normals and therefore light–normal dot products can vary.

D. Its intrinsic albedo must contain every shadow.

Q08 — Multiple choice

What information can a joint RGB histogram retain that three separate channel histograms lose?

A. Which red, green and blue values occur together at the same pixels.

B. The scene's illumination-independent albedo distribution, regardless of camera response.

C. The order of coloured regions across image rows, rather than just their colour counts.

D. The exact spatial location of each pixel having a particular colour.

Q09 — Multiple choice

A binary erosion uses a cross-shaped 3 by 3 structuring element with zeros at its corners. How should the zeros be treated?

A. Require the image's corner positions to be zero.

B. Replace the operation by an average over all nine pixels.

C. Include the mask zeros in a logical AND, forcing every output to zero.

D. Ignore those positions; require foreground only at active element positions.

Q10 — Multiple choice

A system labels every pixel by semantic class, including pixels of several touching people, but gives no person-instance IDs. What does its output provide, and can connected-component labelling alone reliably supply the missing instance separation?

A. Semantic segmentation; no, touching people can merge into one connected component.

B. Instance segmentation; a semantic class label automatically identifies which individual person owns each pixel.

C. Object detection only; connected-component labels are already separate bounding-box detections.

D. Panoptic segmentation; instance IDs are unnecessary as long as all pixels have semantic labels.

Q11 — Multiple choice

Why is a median filter often preferable to averaging for sparse salt-and-pepper noise?

A. I_t is a linear filter with a fixed kernel of average weights.

B. I_t increases the magnitude of all high-frequency components.

C. I_t is less affected by extreme isolated values and can preserve edges better.

D. I_t guarantees removal of every type of noise without information loss.

Q12 — Multiple choice

A Gabor filter is best described as what?

A. A sinusoidal pattern modulated by a Gaussian envelope, selective for local frequency and orientation.

B. An unwindowed global sinusoid that has no spatial localization.

C. A median of oriented derivative responses, making it a nonlinear rank filter.

D. A Gaussian smoothing filter with no sinusoidal carrier and no orientation selectivity.

Q13 — Multiple choice

You multiply two same-length unpadded DFTs and take the inverse DFT. What operation does this normally implement?

A. Pointwise multiplication of the original spatial samples.

B. Circular convolution, with wrap-around at the signal boundaries.

C. Cross-correlation with no complex conjugation or reversal.

D. Full linear convolution of length $N + M - 1$ regardless of padding.

Q14 — Multiple choice

A kernel measures the x derivative of an image. Which edge orientation usually gives a large response?

A. Only a diagonal edge with exactly 45-degree slope.

B. A horizontal edge: intensity changes only across y.

C. A vertical edge: intensity changes across x.

D. A perfectly constant patch.

Q15 — Multiple choice

Compared with a small Gaussian σ, what is a likely effect of increasing σ in Canny?

A. Suppress finer detail and noise, while potentially merging edges and reducing localization precision.

B. Remove the need for non-maximum suppression.

C. Make the operation exactly scale invariant.

D. Guarantee detection of every weak edge with perfect localization.

Q16 — Multiple choice

For a structure tensor at a pixel, which implementation follows the usual order of operations?

A. Compute I_x^2, $I_x I_y$ and I_y^2 first, then aggregate or smooth those products over the neighbourhood.

B. Use I_{xx} and I_{yy} as the diagonal entries and I_{xy} off diagonal, because the structure tensor is the image Hessian.

C. Smooth I_x and I_y to one mean each, then use only the outer product of those two means.

D. Aggregate only I_x and I_y and use their signed means as the diagonal entries, setting the off-diagonal entries to zero.

Q17 — Multiple choice

Shi–Tomasi's corner score for structure-tensor eigenvalues λ_1 and λ_2 is which expression?

A. $\max(\lambda_1, \lambda_2)$

B. $\lambda_1 + \lambda_2$, regardless of their relative sizes

C. $\min(\lambda_1, \lambda_2)$

D. $\lambda_1 - \lambda_2$

Q18 — Multiple choice

A Lucas–Kanade window contains 100 nonzero image-gradient vectors, all parallel. Is its two-component flow uniquely observable from those equations?

A. No; the design matrix has rank 1 and tangential motion remains ambiguous.

B. Yes, because there are more equations than unknowns.

C. No, because least squares works only for exactly two pixels.

D. Yes, because every gradient is nonzero.

Q19 — Multiple choice

Move a flow estimate from a half-resolution pyramid level to full resolution. After spatially resizing the flow arrays, what else is required?

A. Keep the vector magnitudes unchanged because the arrays were resized.

B. Divide both displacement components by 2.

C. Multiply displacement components by 2, then warp and refine at the finer level.

D. Set every displacement to zero before the finer solve.

Q20 — Multiple choice

A fixed image template is tracked with SSD. The target's shape and position remain the same but all intensities gain a constant positive offset. Which claim is correct?

A. SSD can increase even for the true match; a mean-normalized comparison can handle an additive offset better.

B. Unnormalized correlation always removes additive and multiplicative brightness changes.

C. Only the target's geometric rotation can change SSD.

D. SSD is exactly invariant to every additive intensity change.

Q21 — Multiple choice

Which statement about mean-shift clustering is correct?

A. I_t always requires the desired number of clusters K as its central input.

B. I_t minimizes the same fixed-K objective as k-means using identical updates.

C. Its result cannot depend on the bandwidth.

D. I_t seeks density modes; kernel bandwidth influences how many modes and clusters are found.

Q22 — Multiple choice

Retrieve images of the same planar building facade using a query in which part of the facade is occluded by a tree. Assuming enough distinctive features remain visible, which local-keypoint approach is appropriate?

A. Reject every database image unless every query descriptor has a correct facade match, including descriptors on the tree.

B. Accept every nearest descriptor match as geometrically correct, because SIFT's scale robustness guarantees the same scene point.

C. Rank only by each database image's total keypoint count; descriptor appearance and geometric consistency are unnecessary.

D. Match visible local descriptors and verify a consistent homography; the occluded parts need not match.

Q23 — Multiple choice

What is the generic minimum data requirement for an unconstrained 2D affine transformation?

A. Two point pairs because an affine transformation is only rotation plus translation.

B. Four collinear point pairs always uniquely determine it.

C. One point pair because affine transforms have two parameters.

D. Three point pairs whose source points are noncollinear.

Q24 — Multiple choice

Which statement accurately describes backward image warping?

A. I_t guarantees valid source data for every destination pixel, even outside source coverage.

B. I_t forward-maps each source pixel without interpolation.

C. I_t always avoids aliasing during downsampling, even without a low-pass filter.

D. For each destination pixel, inverse-map its coordinate and sample/interpolate the source; out-of-domain samples still need a boundary rule.

Q25 — Multiple choice

What does the SVM kernel trick allow?

A. Evaluate inner products in a transformed feature space without explicitly constructing all transformed features.

B. Make every nonlinear decision boundary linear in the original coordinates.

C. Change the distance metric only after fitting a linear SVM, without changing any training computation.

D. Automatically eliminate every mislabeled training example, making soft-margin penalties unnecessary.

Q26 — Multiple choice

At a fixed detection IoU threshold, two same-class detections both overlap the same ground-truth object sufficiently. In score-ordered, one-to-one matching, how are they counted?

A. Both are TPs but each receives half a recall increment, making this identical to a single detection.

B. The first can be a TP; the second is a duplicate FP unless it can validly match another unmatched object.

C. The first is a TP and the second is ignored automatically in AP, so duplicate boxes cannot reduce precision.

D. Both are TPs because their IoUs are high, even though they claim the same object.

Q27 — Multiple choice

Three same-class boxes have descending scores $A > B > C$. $\mathrm{IoU}(A, B) >$ threshold, $\mathrm{IoU}(B, C) >$ threshold, but $\mathrm{IoU}(A, C) <$ threshold. What does greedy hard NMS keep?

A. A only, because overlap is transitively inherited through B.

B. A and C.

C. B and C.

D. All three, because A and C do not overlap strongly.

Q28 — Multiple choice

Why is identical constant initialization of all interchangeable hidden units problematic?

A. Random initialization guarantees no overfitting.

B. Every nonzero constant necessarily causes exploding gradients on the first step.

C. Biases must always be random for any network to learn at all.

D. Symmetric units can receive identical activations and gradients, so they fail to learn distinct features.

Q29 — Multiple choice

Which statement about Adam and learning rate is correct?

A. Adam eliminates base learning-rate tuning because its moment estimates guarantee stable convergence for any chosen step size.

B. Adam uses gradient-moment estimates to adapt parameter updates, but still has a base learning-rate hyperparameter.

C. Adam uses only a running mean of gradients, so it is exactly SGD with momentum and has no second-moment adaptation.

D. Increasing Adam's base learning rate cannot destabilize training because all adaptive updates are independent of that rate.

Q30 — Multiple choice

A camera only rotates about its optical centre while viewing a static scene. Why is this unsuitable as the sole initialization for ordinary depth triangulation?

A. The pure-rotation homography is invertible, so its inverse already provides each point's unique metric depth.

B. There is no translation baseline and no depth-dependent parallax.

C. Higher image resolution creates a metric translation baseline even when the optical centre stays fixed.

D. A pure rotation makes all otherwise visible feature correspondences invalid.

Part B — Numerical values

Q31 — Numerical

A world-to-camera extrinsic transform is $X_c = RX_w + t$, with

$$R = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$$

and $t = \begin{bmatrix} -3 & 2 & 1 \end{bmatrix}^\top$ metres. Report the Euclidean distance of the camera centre from the world origin. Use $C = -R^\top t$.

Answer: _____

Q32 — Numerical

An RGB image has per-channel means $\begin{bmatrix} 80 & 40 & 20 \end{bmatrix}$. A pixel has RGB value $\begin{bmatrix} 100 & 30 & 10 \end{bmatrix}$. Grey-World correction uses target $= \frac{80+40+20}{3}$, then

$$\text{corrected} = \frac{\text{pixel}}{\text{channel means}} \cdot \text{target}.$$

Ignore clipping and quantization. Report the corrected green value.

Answer: _____

Q33 — Numerical

Weighted photometric stereo minimizes $\sum_j (w_j(L_j g - i_j))^2$. The last light vector's magnitude also encodes its strength. Report g_2.

```
import numpy as np
L = np.array([[1,0,0],[0,1,0],[0,0,1],[1,1,1]], dtype=float)
i = np.array([.2,.3,.6,1.2])
w = np.array([1,1,1,2], dtype=float)
```

Provided routine: `np.linalg.lstsq(L*w[:,None], i*w, rcond=None)[0]` minimizes the stated weighted residuals.

Answer: _____

Q34 — Numerical

Use the simplified Phong model $I = I_a + k_d \max(n \cdot l, 0) + k_s \max(v \cdot r, 0)^s$. Let $I_a = 0.1$, $k_d = 0.6$, $n \cdot l = 0.5$, $k_s = 0.8$, $v \cdot r = 0.5$, and $s = 4$. Report I.

Answer: _____

Q35 — Numerical

Linearly stretch intensities from input range $\begin{bmatrix} 50 & 150 \end{bmatrix}$ to $\begin{bmatrix} 0 & 255 \end{bmatrix}$. Report the output for input intensity 70. No clipping is needed for this value.

Answer: _____

Q36 — Numerical

For the binary image below, compute the number of foreground components under 4-connectivity and under 8-connectivity. Report $\text{count}_4 - \text{count}_8$ as one exact integer. Background is not counted.

$$\begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Answer: _____

Q37 — Numerical

Report the centre output of a 3 by 3 median filter on this patch. Use all nine entries and report an exact integer.

$$\begin{bmatrix} 0 & 7 & 8 \\ 8 & 255 & 9 \\ 10 & 11 & 0 \end{bmatrix}$$

Answer: _____

Q38 — Numerical

Construct a magnitude-weighted UNSIGNED HoG histogram from $G_x = \begin{bmatrix} 3 & -3 & 0 & 2 \end{bmatrix}$ and $G_y = \begin{bmatrix} 4 & -4 & 5 & 0 \end{bmatrix}$. Use angles $\theta = \text{degrees}(\text{atan2}(G_y, G_x)) \bmod 180$. Use nine HARD bins $[0,20)$,$[20, 40)$,...,$[160,180)$, numbered 0..8; no interpolation and no histogram normalization. Report the value of bin 2, which covers $[40,60)$ degrees.
Provided functions: `np.arctan2(y,x)` gives quadrant-aware angles in radians; `np.degrees(angle)` converts radians to degrees. Magnitude is `np.sqrt(Gx*Gx+Gy*Gy)`.

Answer: _____

Q39 — Numerical

A complex Fourier coefficient is $F = -3 - 4\text{i}$. Report its principal phase in degrees in the range (-180, 180]. Provided function: `np.degrees(np.arctan2(F.imag, F.real))` uses quadrant-aware atan2(y, x).

Answer: _____

Q40 — Numerical

In Lucas–Kanade, solve $A \begin{bmatrix} u & v \end{bmatrix}^{\mathsf{T}} = -I_t$ by least squares and report v.

```
import numpy as np
A = np.array([[2,1],[1,-1],[0,2],[3,0]], dtype=float)
It = np.array([-5,-1,-3,-6], dtype=float)
```

Provided routine: `np.linalg.lstsq(A, -It, rcond=None)[0]` returns [u,v].

Answer: _____

Q41 — Numerical

Report the zero-based index returned by this SSD matching code. NumPy's argmin returns the FIRST minimum in a tie.

```
import numpy as np
T = np.array([[1,2],[3,4]])
P = np.array([[[2,3],[4,5]],
              [[1,1],[3,5]],
              [[2,2],[2,4]]])
costs = ((P-T)**2).sum(axis=(1,2))
answer = int(np.argmin(costs))
```

Answer: _____

Q42 — Numerical

Fit a 2D affine transform by least squares. Source rows are

$$\begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix};$$

target rows are

$$\begin{bmatrix} 2 & -1 \\ 4 & -1 \\ 3 & 2 \\ 5 & 3 \end{bmatrix}.$$

Build A with rows $[x, y, 1]$ and solve $AP =$ targets. Report the predicted target y coordinate for source (2,3).
Provided routine: `P=np.linalg.lstsq(A, targets, rcond=None)[0]`; prediction is `np.array([2,3,1])@P`.

Answer: _____

Q43 — Numerical

Rows contain [nearest_distance, second_nearest_distance] from descriptor matching. Keep a match iff $\frac{d_1}{d_2} < 0.75$ (strict inequality). Report the count kept.

$$\begin{bmatrix} 0.3 & 0.5 \\ 0.4 & 0.5 \\ 0.6 & 1.0 \\ 0.2 & 0.21 \\ 0.1 & 0.4 \end{bmatrix}$$

Answer: _____

Q44 — Numerical

Run ONE assignment/update iteration of 1D k-means with initial centres $\begin{bmatrix} 0 & 10 \end{bmatrix}$ and data $\begin{bmatrix} 0 & 2 & 3 & 9 & 11 \end{bmatrix}$. Use squared Euclidean distance. Report the updated smaller centre.

Answer: _____

Q45 — Numerical

Rows are actual classes; columns are predicted classes. Report MACRO recall as a fraction for this matrix.

$$\begin{bmatrix} 30 & 10 & 0 \\ 5 & 10 & 5 \\ 0 & 2 & 8 \end{bmatrix}$$

Answer: _____

Q46 — Numerical

Greedy hard NMS processes one class in score order A,B,C,D (all scores distinct). Exact pairwise IoUs are below; fractions mean ordinary division. Suppress a lower-scoring box only when its IoU with a KEPT box is strictly greater than 0.5. Report the number kept.

	A	B	C	D
A	1	$\frac{3}{5}$	$\frac{1}{3}$	$\frac{1}{7}$
B	$\frac{3}{5}$	1	$\frac{3}{5}$	$\frac{1}{3}$
C	$\frac{1}{3}$	$\frac{3}{5}$	1	$\frac{3}{5}$
D	$\frac{1}{7}$	$\frac{1}{3}$	$\frac{3}{5}$	1

Answer: _____

Q47 — Numerical

A CNN layer uses a valid stride-1 CROSS-CORRELATION, no bias,

$$I = \begin{bmatrix} 1 & 3 & 5 \\ 2 & -1 & 4 \end{bmatrix}$$

and

$$W = \begin{bmatrix} 1 & 0 \\ -1 & 2 \end{bmatrix}.$$

Its two output positions receive upstream loss gradients $\begin{bmatrix} 2 & -1 \end{bmatrix}$. Report $\frac{\partial \text{Loss}}{\partial W_{1,1}}$, with zero-based indices. The derivative sums upstream gradient · the corresponding input value over BOTH output positions.

Answer: _____

Q48 — Numerical

A final feature map has shape 9×9×8. Global average pooling replaces each channel by its spatial mean. A fully connected 5-class head then uses one bias per class. Report the head's trainable parameter count; exclude the backbone.

Answer: _____

Q49 — Numerical

Use ORIGINAL YOLO v1, whose output shape is $S \times S \times (5B + C)$: B boxes with 5 values each, and ONE shared C-class probability vector per cell. Let $S = 7$, $B = 2$ and $C = 4$. Report the total number of scalar outputs.

Answer: _____

Q50 — Numerical

An ideal rectified stereo pair has focal length $f = 1200$ pixels, baseline $B = 0.12$ metres, and matching horizontal coordinates $u_L = 414,\ u_R = 390$. Both principal points are equal. Use disparity $d = u_L - u_R$ and depth $Z = \frac{fB}{d}$. Report Z in metres.

Answer: _____